Year	Annual Returns (%)		
	Fund ABC	Fund XYZ	Fund PQR
Mean	−4.0	−10.8	−5.0
Standard deviation	17.8	15.6	10.5

2. The fund with the mean absolute deviation (MAD) is Fund:

   A. ABC.

   B. XYZ.

   C. PQR.

   Solution:

   A is correct. The MAD of Fund ABC's returns is the highest among the three funds. Using Equation 3, the MAD for each fund is calculated as follows:

   MAD for Fund ABC =

   $$\frac{|-20-(-4)|+|23-(-4)|+|-14-(-4)|+|5-(-4)|+|-14-(-4)|}{5} = 14.4\%.$$

   MAD for Fund XYZ =

   $$\frac{|-33-(-10.8)|+|-12-(-10.8)|+|-12-(-10.8)|+|-8-(-10.8)|+|11-(-10.8)|}{5}$$

   $= 9.8\%.$

   MAD for Fund PQR =

   $$\frac{|-14-(-5)|+|-18-(-5)|+|6-(-5)|+|-2-(-5)|+|3-(-5)|}{5} = 8.8\%.$$

3. Consider the statistics in Exhibit 15 for Portfolio A and Portfolio B over the past 12 months:

Exhibit 15: Portfolio A and Portfolio B Statistics

	Portfolio A	Portfolio B
Average Return	3%	3%
Geometric Return	2.85%	?
Standard Deviation	4%	6%

4. The geometric mean return of Portfolio B is *most likely* to be:

   A. less than 2.85 percent.

   B. equal to 2.85 percent.

   C. greater than 2.85 percent.

   Solution:

   A is correct. The higher the dispersion of a distribution, the greater the difference between the arithmetic mean and the geometric mean.